Babäk Firoozi Fooladi

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Real Estate Economist | Data Analyst | Data Engineer

I am a PhD candidate in real estate economics at Aalto University, possessing a robust foundation in data science, econometrics, and machine learning. I have over five years of experience in delivering end-to-end data engineering and analytics solutions across academia, consulting, and industry. My work ranges from data pipelines and spatial data infrastructure to employing advanced statistical and machine learning methods for research, analysis and providing insights into complex data. I enjoy and grow in collaborative, cross-disciplinary environments where questions and problems are diverse and more engaging.

Soft Skills

Stakeholde management:	●●●●	**Visual communication:**	●●●●	**Task management:**	●●●●
Team work:	●●●	**Self-organised:**	●●●	**Project management:**	●●●●
Accounting:	●●	**Quality control:**	●●●●	**Graphic Design:**	●●●●●
Quantiative research:	●●●●	**Prompt engineering:**	●●●●	**Document Management:**	●●●●

Technical Skills

Python:	●●●	**PostgreSQL:**	●●●	**DuckDB:**	●●●●	**SQL:**	●●●
Spatial data:	●●●●●	**LLM/NLP/LLM:**	●●●●	**SQLite:**	●●●	**AZURE:**	●●
GCP:	●●●●	**R:**	●●●●	**MongoDB:**	●●	**Excel:**	●●●●
AcrGIS:	●●●	**ETL/ELT:**	●●●●	**Cartography:**	●●●●	**STATA:**	●●●●
Git:	●●●	**RAG:**	●●●●●	**Julia:**	●●	**PowerBI:**	●●

Education

Aalto University — Espoo, Finland
Doctor of Philosophy in Technology, Real Estate Economics — *Sep 2020 - Present*

Aalto University — Espoo, Finland
MSc. (Tech.) Urban Studies and Planning in Real Estate Economic — *Sep 2018 - Jul 2020*
GPA: 4.1/5, Graduated with honours

University of Tehran — Tehran, Iran
BA. Town Planning — *Sep 2011 - Jun 2015*
GPA: 16.67/20, Academic excellence

Experience

DATA ANALYTICS & ENGINEERING — Jan 2025 – Present
Self-Employed | Consultant — *Project-Based – Espoo, Finland*
Delivering end-to-end data engineering, analytics, and BI solutions tailored to each client's industry and business objectives. Advising on data strategy with a focus on scalability and sustainable growth. As a data specialist, I design efficient pipelines that handle diverse data types and translate complex data into clear, actionable insights aligned with client goals.

Building Information Backend Jan 2026 – Present

Apex-Heat – Finland

Stakeholders asked for spatial data on all buildings in Finland that could be used in other EU countries. The main problem was figuring out what data was needed and how to get it and put it all together. I developed the scripts to acquire data from official sources, process them and correct their goemetry, and then compute the key metrics that the stakeholders needed. I also designed and implemented an API-based pipeline to automate the data operations in Finland, and evaluated and structured the spatial data models for EU-wide coverage. I also modelled the data leveraging my domain expertise in real estate economics and zoning regulations.

- Geopandas, Shapely, Geoparquet, DuckDB, PostGIS
- Developing a data pipeline to transform and compute key metrics from building spatial data across Finland.
- Designing and implementing an **API-based pipeline** for automated data operations in Finland.
- Evaluating and structuring spatial data models for EU-wide coverage.
- Modelling data leveraging domain expertise in real estate economics and zoning regulations.

Optimisation & Technical Debt Reduction Jan 2025 – Dec 2025

Apex-Heat – Finland

Led the move of core application calculations to Polars to make them run faster, while also taking care of some technical debt issues, such as writing down how to use key features. To keep things in check, the team only worked on a small set of calculations, but they had a clear plan for how to move the rest of the components in future iterations. The work led to an 86% improvement in the targeted calculations and a 60% improvement across the whole application.

- Identified and resolved **technical debt**, significantly improving data pipeline performance and maintainability.
- Led migration from **Pandas** to **Polars** (including GeoPandas)
- Refactored codebase design patterns and produced **technical documentation**.
- Expanded and improved **test coverage** and testing methodologies for data processing workflows.

Ph.D. Candidate Sep 2020 – Present

Aalto University *Full-time – Espoo, Finland*

Performing doctoral research at the crossroads of urban economics, urban planning and real estate, using econometric and machine learning methods on housing markets, real estate development, and city planning. In charge of overseeing the entire spatial data lifecycle for research projects, from gathering and processing the data to analyzing and visualizing it. Also responsible for building and maintaining the team's data infrastructure. Created and set up a data lake with Hive partitioning for research datasets. Also made DuckDB-based extraction workflows with easy-to-understand documentation so that researchers who aren't tech-savvy can query and use the data on their own.

- Preparing and maintaining **data visualizations** for academic presentations and lectures.
- Designing data-pipelines for local AI to extract features from text.
- Applying Econometric, ML, time-series and data analysis.
- Acquiring GIS data from REST and ArcGIS Online endpoints.
- Deploying and maintaining **Azure Functions** to automate web scraping of property listings and store real estate data for research purposes.
- Performing **GIS analysis** to produce and maintain geospatial research datasets and geocode real estate transactions from data sources such as MML, Ryhti, SYKE, and KVKL.
- Designing and deploying research databases using **PostgreSQL**, **DuckDB**, and data **hive partitions**
- Data stewardship responsibilities to acquire, curate, and distribute research datasets.

Research Assistant Jun 2019 – Sep 2019

Aalto University Full-time – Espoo, Finland

Contributed to an **Agent-Based Modelling (ABM)** research project simulating urban mobility and parking pressure, developed in collaboration with **MIT CityLab**.

- Developed an **ABM simulation** to model parking pressure dynamics in the Otaniemi campus area.
- Collaborated with **MIT CityLab** on agent-based modelling of urban mobility patterns.

- Implemented simulations using the **GAMA Platform**.

Data Analytics Engineer — Sep 2020 – Dec 2024
Qissa kaupunkisuunnitteluanalytiikka Oy – Startup — *Part-time – Espoo, Finland*

Delivered data engineering and spatial analysis solutions for an urban analytics startup, supporting clients in urban planning and sustainability with end-to-end data workflows.

- Designed and maintained data pipelines to ingest **GIS data** from REST APIs, performing cleaning, **geocoding**, and storage in line with client specifications.
- Built a **spatial analytics dashboard** with a network analysis backend, integrating **HSL, HERE, and HSY APIs** to gather, process, and perform **spatial network analysis**.
- Extracted, processed, and served **census data** from the US, Denmark, Sweden, and Finland to calculate CO_2 emissions from commuting patterns.

Planning Assistant — Nov 2018 – Jun 2021
WSP Finland Oy — *Part-time – Helsinki, Finland*

Provided GIS and spatial data analysis support across urban planning and transport projects, delivering actionable insights and visualizations to multidisciplinary teams and external clients.

- Conducted **spatial data analysis**, processing and managing geospatial datasets across multiple concurrent projects.
- Performed **pedestrian network analysis** to forecast route choice and model post-development pedestrian flows, calibrated against real-world conditions.
- Prepared maps, graphs, and **data visualizations** for stakeholders, project managers, and clients to support planning decisions.
- Delivered **GIS analysis** to cross-disciplinary teams including transport engineers, landscape designers, urban planners, and architects.
- Configured **ArcGIS Online** platforms to support **maritime spatial planning** workflows.
- Developed dynamic **3D city models** using **ESRI CityEngine** to visualize urban development scenarios.

GIS Operator — Aug 2015 – May 2016
University of Tehran — *Part-time – Tehran, Iran*

- Collected, processed, and prepared **GIS datasets** to support urban designers in implementing and evaluating new urban design directives for Tehran.

Machine Learning | Statistical models | Econometrics

Uni-variate time-series: ●●●	**VAR time-series:** ●●●	**Co-integration time-series:** ●●
multi-variate time-series: ●●●	**OLS:** ●●●●●	**IV-OLS:** ●●●●
Panel data: ●●●●	**Generalised Method of Moments:** ●●●●	**Generalised linear models:** ●●
Spatial econometrics: ●●●●	**Survival/time-to-event:** ●●●●●	**Additive hazard models:** ●●●●●

Volunteer Work

- DataTribe collective moderator, contributor
- DataTribe collective speaker: Spatial data analysis, spatial data engineering
- Helsinki Data Week volunteer (2024 & 2025), contributor